

Full INS/GNSS Reference Filter Mathematical Formulation

sensor_fusion documentation

May 8, 2026

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1 Scope

This document describes the Full INS/GNSS filter implemented in `sensor_fusion/src/full/` and `sensor_fusion/src/full/generated.rs`. It is used for replay, diagnostics, and comparison with the runtime Reduced. It carries ECEF navigation states, IMU bias and scale correction states, and the physical vehicle-to-body mount used by inertial propagation.

2 Frames and Conventions

Symbol	Meaning
e	ECEF frame. Full nominal position and velocity are propagated in this frame.
n	Local NED frame. GNSS standard deviations may be supplied in this frame and are rotated into ECEF.
b	Raw IMU body/sensor frame. Full IMU deltas are supplied in this frame.
v	Vehicle frame, forward-right-down. NHC is expressed in this frame.

Rotations follow the project convention

$$x_a = C_{ab}x_b, \quad C(q_1 \otimes q_2) = C(q_1)C(q_2). \quad (1)$$

The full attitude quaternion q_{ev} maps vehicle-frame vectors into ECEF:

$$x_e = C_{ev}(q_{ev})x_v. \quad (2)$$

The mount quaternion stored in the Rust fields `qcs0..qcs3` is the physical vehicle-to-body mount q_{bv} :

$$x_b = C_{bv}(q_{bv})x_v, \quad x_v = C_{bv}(q_{bv})^T x_b. \quad (3)$$

The local NED-to-ECEF relation is represented by

$$x_n = C_{ne}x_e, \quad x_e = C_{ne}^T x_n. \quad (4)$$

3 Nominal State

The nominal state contains 26 scalar fields:

$$x = \begin{bmatrix} q_{ev} & v_e^T & p_e^T & b_g^T & b_a^T & s_g^T & s_a^T & q_{bv} \end{bmatrix}^T. \quad (5)$$

The public Rust fields are:

Fields	Meaning
<code>q0..q3</code>	q_{ev} , vehicle frame to ECEF attitude
<code>vn,ve,vd</code>	ECEF velocity v_e , meters per second
<code>pn,pe,pd</code>	ECEF position p_e , meters
<code>bgx..bgz</code>	gyro additive correction b_g , radians per second
<code>bax..baz</code>	accelerometer additive correction b_a , meters per second squared
<code>sgx..sgz</code>	gyro scale correction s_g
<code>sax..saz</code>	accelerometer scale correction s_a
<code>qcs0..qcs3</code>	physical vehicle-to-body mount quaternion q_{bv}

The code applies scale and additive correction directly to IMU rates and specific force:

$$\omega_b^c = s_g \odot \omega_{\text{raw},b} + b_g, \quad (6)$$

$$f_b^c = s_a \odot f_{\text{raw},b} + b_a. \quad (7)$$

Consequently, these are correction states. A physical sensor bias appears with the opposite sign of the additive correction required to remove it.

4 Error State

The 24-dimensional error state order in the Rust update and generated matrices is

$$\delta x = \begin{bmatrix} \delta p_e & \delta v_e & \delta \theta_v & \delta b_a & \delta b_g & \delta s_a & \delta s_g & \delta \psi_{bv} \end{bmatrix}^T. \quad (8)$$

Index ranges are:

Indices	Error component
0..2	ECEF position error δp_e
3..5	ECEF velocity error δv_e
6..8	vehicle-attitude small angle $\delta \theta_v$
9..11	accelerometer-bias correction error δb_a
12..14	gyro-bias correction error δb_g
15..17	accelerometer-scale error δs_a
18..20	gyro-scale error δs_g
21..23	vehicle-to-body mount error $\delta \psi_{bv}$

5 Nominal Propagation

The full filter uses a two-sample IMU increment. For each half-step $i \in \{1, 2\}$,

$$\omega_{b,i}^c = s_g \odot \left(\frac{\Delta \theta_{b,i}}{\Delta t} \right) + b_g, \quad (9)$$

$$f_{b,i}^c = s_a \odot \left(\frac{\Delta v_{b,i}}{\Delta t} \right) + b_a, \quad (10)$$

$$\omega_{v,i} = C_{bv}(q_{bv})^T \omega_{b,i}^c, \quad (11)$$

$$f_{v,i} = C_{bv}(q_{bv})^T f_{b,i}^c. \quad (12)$$

The attitude derivative includes Earth rotation:

$$\dot{q}_{ev} = \frac{1}{2} q_{ev} \otimes \begin{bmatrix} 0 & \omega_v^T \end{bmatrix}^T + \frac{1}{2} \begin{bmatrix} \omega_{ie} q_3 & \omega_{ie} q_2 & -\omega_{ie} q_1 & -\omega_{ie} q_0 \end{bmatrix}^T. \quad (13)$$

Specific force is rotated into ECEF:

$$f_e = C_{ev} f_v. \quad (14)$$

Velocity dynamics use the J2 ECEF gravity model and Coriolis terms:

$$\dot{v}_e = f_e + g_e(p_e) - 2 \Omega_{ie}^e \times v_e. \quad (15)$$

The implementation writes this as component terms using $\omega_{ie} = 7.292115 \times 10^{-5}$ rad/s. Position obeys

$$\dot{p}_e = v_e. \quad (16)$$

The nominal propagation uses a predictor-corrector/trapezoidal structure: evaluate derivatives at the current state, form a temporary state after Δt , evaluate derivatives again, then average the two derivatives for the final position, velocity, and quaternion update. The nominal bias, scale, and mount states are constant during propagation.

6 Error Propagation

The generated full model provides

$$\delta x_{k+1} = F_k \delta x_k + G_k w_k, \quad P_{k+1}^- = F_k P_k^+ F_k^T + G_k Q_k G_k^T. \quad (17)$$

The 21-noise vector groups are accelerometer white noise, gyro white noise, accelerometer-bias random walk, gyro-bias random walk, accelerometer-scale random walk, gyro-scale random walk, and mount random walk:

$$w = \begin{bmatrix} n_a & n_g & n_{ba} & n_{bg} & n_{sa} & n_{sg} & n_m \end{bmatrix}^T. \quad (18)$$

The runtime diagonal entries are

$$Q_k = \text{diag} \left(\sigma_a^2 \Delta t I_3, \sigma_g^2 \Delta t I_3, \sigma_{ba}^2 \Delta t I_3, \sigma_{bg}^2 \Delta t I_3, \sigma_{sa}^2 \Delta t I_3, \sigma_{sg}^2 \Delta t I_3, \sigma_m^2 \Delta t I_3 \right). \quad (19)$$

7 GNSS Reference Measurements

7.1 Position

GNSS position is supplied in ECEF:

$$z_p = p_e + \eta_p. \quad (20)$$

When a horizontal accuracy h_{acc} is supplied, the implementation forms a local NED diagonal covariance

$$R_n = \text{diag} \left(h_{\text{acc}}^2, h_{\text{acc}}^2, (2.5h_{\text{acc}})^2 \right), \quad (21)$$

rotates it to ECEF, then applies a Cholesky whitening transform T_p . The actual rows fused are

$$r_p = T_p(z_p - \hat{p}_e), \quad H_p = T_p \begin{bmatrix} I_3 & 0 & \cdots & 0 \end{bmatrix}. \quad (22)$$

7.2 Velocity

GNSS velocity is supplied in ECEF, with either a scalar speed accuracy or per-axis NED velocity standard deviations. In the per-axis case, the NED velocity covariance is rotated to ECEF and Cholesky-whitened:

$$r_v = T_v(z_v - \hat{v}_e), \quad H_v = T_v \begin{bmatrix} 0 & I_3 & 0 & \cdots & 0 \end{bmatrix}. \quad (23)$$

If only scalar speed accuracy is available, the three ECEF velocity rows use that scalar variance directly.

8 Non-Holonomic Constraint

The full NHC predicts vehicle-frame velocity from ECEF velocity:

$$v_v = C_{ev}^T v_e. \quad (24)$$

The lateral and vertical constraints are

$$h_y = e_y^T v_v \approx 0, \quad h_z = e_z^T v_v \approx 0. \quad (25)$$

The generated rows include derivatives with respect to velocity and vehicle attitude. They have no direct mount block because mount affects NHC through propagation-induced covariance, not through the instantaneous NHC measurement:

$$\frac{\partial(e_i^T v_v)}{\partial \delta \psi_{bv}} = 0, \quad i \in \{y, z\}. \quad (26)$$

NHC rows are gated by speed and a quasi-static IMU check:

$$\|v_e\| \geq 0.5 \text{ m/s}, \quad \|\omega_b\| < 0.2 \text{ rad/s}, \quad |||f_b|| - 9.81| < 1.0 \text{ m/s}^2. \quad (27)$$

A scalar chi-square gate prevents large residuals from being fused.

9 Batch Update and Injection

The full filter batches up to eight rows: three position rows, three velocity rows, and two NHC rows. It uses a sequential covariance update in f64 shadow storage:

$$S_i = H_i P H_i^T + R_i, \quad (28)$$

$$K_i = P H_i^T S_i^{-1}, \quad (29)$$

$$\delta x \leftarrow \delta x + K_i (r_i - H_i \delta x), \quad (30)$$

$$P \leftarrow P - K_i S_i K_i^T. \quad (31)$$

After the batch, the accumulated error is injected:

$$p_e^+ = p_e^- + \delta p_e, \quad v_e^+ = v_e^- + \delta v_e, \quad (32)$$

$$q_{ev}^+ = \delta q(\delta \theta_v) \otimes q_{ev}^-, \quad (33)$$

$$b_a^+ = b_a^- + \delta b_a, \quad b_g^+ = b_g^- + \delta b_g, \quad (34)$$

$$s_a^+ = s_a^- + \delta s_a, \quad s_g^+ = s_g^- + \delta s_g, \quad (35)$$

$$q_{bv}^+ = \delta q(\delta \psi_{bv}) \otimes q_{bv}^-. \quad (36)$$

Both quaternions are normalized. After injection, the attitude and mount covariance blocks are reset with the generated first-order quaternion reset Jacobian

$$G_r(\delta \theta) \simeq I - \frac{1}{2} [\delta \theta]_{\times}. \quad (37)$$

The reset is applied to the q_{ev} attitude block and to the q_{bv} mount block, then the covariance is symmetrized.

10 ECEF/NED Initialization

For visualizer replay, a vehicle yaw in the local NED frame is converted to vehicle-to-ECEF attitude q_{ev} using latitude and longitude. The physical mount q_{bv} is then initialized from Align, reference data, or a manual caller-supplied mount. Raw body-frame IMU samples are supplied directly to the Full filter.

11 Observability and Known Limitations

- GNSS position and velocity directly observe only ECEF position and velocity. Attitude, biases, scales, and mount are updated through covariance coupling and inertial dynamics.
- Lateral and vertical NHC rows do not directly observe mount. Mount is updated through propagation-induced covariance with attitude and velocity.
- Accelerometer vertical bias and accelerometer vertical scale are strongly coupled in ground-vehicle motion dominated by gravity. The visualizer commonly keeps accelerometer scale fixed for diagnostics.
- The additive bias states are correction states. Diagnostic plots should not be interpreted as physical sensor bias without applying the sign convention.
- Mount, attitude, gyro bias, and gyro scale can become correlated during turns. Compare vehicle attitude and physical mount separately when diagnosing mount convergence.